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MILITARY SPECIFICATION

BOOSTER, M123

This specification has been approved by the Department of Defense and is mandatory for use by the Departments of the Army, the Navy, and the Air Force.

1. SCOPE

1.1 This specification covers the M123 Booster for use in fuzes for 20-mm ammunition.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, standards, and drawings of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL–A–2550 — Ammunition and Special Weapons, General Specification for.

MIL–D–46582 — Detonator, Stab, T92E5.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–109 — Inspection Terms and Definitions.

MIL–STD–129 — Marking for Shipment and Storage.

DRAWINGS

ORDNANCE CORPS

D7258843 — Booster, M123, Loading Assembly.

D7259406 — Fixture, Booster Output Testing.

C7259407 —Machine, Detonator Testing.

F7548103 — Box, Setup, Packing, Amm, Outer, for Booster, Detonators or Primers. Assembly, Details, Packing and Marking.

F7548105 — Box Packing, Amm for Detonators M37 or M42 or Boosters M123 Assembly, Details and Packing.

FSC 1305

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F8796522 — Marking Diagram and Sealing for Wood Packing Boxes.

PUBLICATIONS

ORDNANCE CORPS

GL–7555878 — Index to Lists of Inspection Gages and Parts Gaged on: Booster, M123, Loading Assembly.

ORD–M608–11 — Procedures and Tables for Continuous Sampling by Attributes.

(Copies of specifications, standards, drawings, and publications required by the contractor in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein.

INTERSTATE COMMERCE COMMISSION

49CFR 71–78 —Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles.

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition—Revised 1950), available from the Superintendent of Documents, Government Printing Office, Washington 25, D.C. Orders for the above publication should cite "49 CFR 71-78 (Rev. 1950).")

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with their applicable specifications.

3.2 Booster assembly. The loaded booster shall comply with all the requirements of its applicable drawings and with all the requirements of the applicable specifications.

3.3 Output. The booster, when initiated by a T92E5 detonator shall have an output sufficient to create a cavity of .085 inch in depth in a sintered brass disc of .500 inch nominal thickness. The method of test will be as specified in 4.5.1.

3.4 Detonator deterioration. The detonators furnished for use in the test specified in 4.5.1 shall be periodically tested for possible deterioration. (see 6.3).

3.5 Preproduction sample. Preproduction samples, as specified in 4.3, shall be required prior to the start of regular production.

3.6 Workmanship. The loaded booster and its components shall be free from cracks, splits, ruptures, perforations, wrinkles, folds, burrs, exposed explosive or other injurious defects. All parts shall be clean and free of foreign matter. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents used.

4. QUALITY ASSURANCE PROVISIONS

4.1 The supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 The provisions of Specification MIL–A–2550 apply to this specification. Reference will be made to MIL–STD–109 for definition of inspection terms used herein.

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4.3 Preproduction samples. Prior to the start of regular production, the contractor shall submit preproduction samples manufactured by the same methods and equipment that will be utilized in performing the work under contract. The contracting officer will furnish instructions for shipment of the sample to a designated Government laboratory for examination and test to ascertain conformance with the requirements of the applicable drawing and of this specification. The preproduction samples shall consist of the following:

(a) 50 booster cups for visual and dimensional inspection.

(b) 100 complete booster assemblies for visual and dimensional inspection.

(c) 150 complete booster assemblies for output test.

4.3.1 Results of the examination and test will be furnished to the contractor. Approval or disapproval of the preproduction sample will be given by the responsible procuring activity. Instructions for the submission of new samples in case of disapproval of the original sample will be issued by the procuring activity.

4.4 Sampling.

4.4.1 *Lot formation.* A lot shall consist of boosters loaded and assembled by only one manufacturer in accordance with the same drawing, same drawing revision, same specification and specification revision. Each booster lot shall contain RDX procured from a single source.

4.4.2 Examination.

4.4.2.1 Each booster assembly of the lot shall be examined by the contractor for the critical defects listed in 4.4.4 and any booster assembly found to be defective, shall be removed from the lot.

4.4.2.2 Lot sampling and acceptance criteria will be in accordance with MIL–STD–105 or ORD–M608–11, as appropriate. Judgment for lot acceptance will be based on either the individual defect characteristic or on classes of defect characteristics with individual defect characteristic limitations, as approved by the procuring activity prior to the initiation of production, with the following AQL's assigned (see 6.2):

4.4.2.2.1 *Individual defect characteristic basis.*

Major defects	0.25%
Minor defects	0.40%

4.4.2.2.2 *Defect characteristic class basis.*

Major defects	1.50%
Minor defects	2 50%

Individual defect characteristic AQL's associated with the class basis will be:

Major defects	0.40
Minor defects	0.65

4.4.3 *Test inspection.* The test specified below will be conducted in accordance with the method specified in 4.5. Acceptance criteria are as specified below:

4.4.3.1 *Output.* A double sampling plan using the method of MIL–STD–105 will be used in this test. First sample size will be fifty (50) boosters and second sample size will be one hundred (100) boosters with acceptance numbers of zero (0) and two (2) respectively. The lot will be finally rejected if three or more boosters in either the first test or in the combined results of the first test and retest fail to meet the requirement of 3.3.

4.4.4 *Classification of defects.*

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4.4.4.1 *Cup Booster (see Drawing C7258844, a detail of Dwg. D7258843).*

Categories and defects	*Method of inspection*
Critical—none defined.	
Major	
101. Foreign matter inside	Visual
102. Length	Gage
Minor	
201. Thickness of base	Gage
202. Outside diameter	Gage
203. Inside diameter	Gage

4.4.4.2 *Booster assembly (see Dwg. D7258843).*

Categories and defects	*Method of inspection*
Critical	
1. Explosive exposed or on exterior	Visual (see 4.4.5.2).
2. Metal defective	Visual (see 4.4.5.1).
3. Disc missing	Visual
Major	
101. Identification color missing or misplaced	Visual (see 4.4.5.3).
102. Outside diameter-max	Gage
103. Length, max	Gage
Minor	
201. Length, min	Gage
202. Outside diameter, min	Gage
205. Crimp not pressed firmly on inner edge for 360°	Visual

4.4.4.3 *Sealed container (see Dwg. F7548103).*

Categories and defects	*Method of inspection*
Critical—none defined.	
Major	
101. Not completely sealed or wrapped	Visual
102. Contents loose	Visual
103. Inner box(es) exposed	Visual
104. Outer pack damaged	Visual
105. Label or alternative identification missing	Visual
106. Incorrect or unidentifiable lot number or loading date	Visual
Minor	
201. Marking other than lot number or loading date incorrect or unidentifiable	Visual

4.4.4.4 *Sealed wooden packing box (see Dwg. F7548105).*

Categories and defects	*Method of inspection*
Critical—none defined.	
Major	
101. Board broken or split	Visual
102. Box damaged to extent that sealed containers are, or are liable to be exposed	Visual
103. More than one strap missing	Visual
104. Lot number or FSN missing, incorrect, incomplete or unidentifiable	Visual
Minor	
201. Contents loose	Manual
202. Box improperly assembled	Visual
203. Marking other than lot number and FSN missing, incorrect, incomplete or unidentifiable	Visual
204. Marking touched up or repaired with nonpermissible material (crayon, chalk, etc.)	Visual

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4.4.5 *Defect standards.*

4.4.5.1 Any crack, split, rupture or perforation in either the booster cup or disc will be considered a defect.

4.4.5.2 The presence of explosive on any outside surface, or exposed explosive around the periphery of the crimp at the junction of the disc, will be considered a defect.

4.4.5.3 The booster will be classed defective if the color code is missing or appears at the wrong end of the booster.

4.4.6 *Inspection equipment.*

4.4.6.1 Government verification inspection will be accomplished with the use of gages and equipment specified in the "Index to Lists of Inspection Gages and Parts Gaged On: Booster, M123, Loading Assembly," GL 7555878.

4.5 Test method.

4.5.1 *Output.* Boosters will be tested for output using the setup and equipment specified on Drawings D7259406 and C7259407 or other equipment approved by the procuring activity. Detonators loaded and assembled in accordance with MIL–D–46582(ORD) will be used to initiate the booster. The 7.96 oz. ball will be dropped upon the firing pin from a height of 10 inches. Sample detonators which fail to function will be replaced without penalty to the booster lot. (see 6.3)

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Level C.* Boosters shall be packaged in accordance with Drawing F7548103.

5.2 Packing.

5.2.1 *Level C.* Boosters packaged as specified in 5.1.1 shall be packed in a wood box in accordance with Drawing F7548105.

5.3 Marking.

5.3.1 *Level C.* Packaged and packed boosters shall be marked in accordance with Drawings F7548103 and F8796522, respectively. All marking shall be in accordance with ICC Regulation 49 CFR 71–78 and with MIL–STD–129 to the extent applicable.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Place of inspection, if not place of manufacture.

6.2 AQL's. The optional use of AQL values for either individual defects or classes of defects, as approved by the procuring activity prior to the initiation of production, is introduced to minimize inspection agency administrative burden which might occur from an exclusive assignment of individual defect AQL's. The option also permits flexibility where sampling inspection for acceptance is integrated into the process.

6.3 Detonator deterioration. Prior to use in testing the first lot of boosters and at least every 90 days thereafter, the detonator lot assigned for use in the booster output testing will be tested for possible deterioration. The tests will be conducted in accordance with MIL–D–46582(ORD) and the detonators must meet the acceptance criteria of MIL–D–46582(ORD). Detonator lots failing to meet the acceptance criteria of MIL–D–46582(ORD) will be referred to the procuring activity and a new lot of detonators will be assigned for use in booster testing.

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